

Session 4: Software Development Lifecycle (SDLC) Explain SDLC phases: Requirement Design Development Testing Deployment Maintenance. Discuss Agile vs Waterfall, Scrum roles (PO, Scrum Master, Dev Team). Relate each phase to a real-world app (e.g., "Spotify new feature rollout"). Trainer Demo: Show a Jira board mock-up with stories in "To Do / In Progress / Done."

TASKS:

1. List all 6 phases of SDLC in order with one line about each.

- **Planning and Requirement Analysis:**
Gather and analyze the needs and expectations of users and stakeholders.
- **System Design:**
Create the system architecture, database structure, and overall design.
- **Development (Coding):**
Write the program code based on the design specifications.
- **Testing:**
Verify that the software works correctly and identify any bugs or errors.
- **Deployment**
Release and install the software for users in the production environment.
- **Maintenance**
Fix issues, update features, and improve the software after deployment.

2. Write 3 differences between Agile and Waterfall models.

Agile:

- Agile follows an iterative and flexible approach.
- Requirements can change during development.
- Testing is done continuously throughout the project.

Waterfall:

- Waterfall follows a linear and sequential approach.
- Requirements are fixed before development starts.
- Testing is done after the development phase is completed.

3. Name the 3 main Scrum roles and write one line about the responsibility of each.

1. Product Owner
2. Scrum Master
3. Development Team

1. Product Owner: Defines product requirements and prioritizes the work to maximize business value.

2. Scrum Master: Facilitates the Scrum process, removes obstacles, and helps the team follow Scrum practices.

3. Development Team: Designs, develops, tests, and delivers the product increment during each sprint.

4. What is a Sprint in Scrum? Mention its typical duration.

A Sprint is a fixed time period in Scrum during which the team completes a set of planned tasks and delivers a working product increment.

Typical Duration: A Sprint usually lasts 1 to 4 weeks, with 2 weeks being the most common duration.

5. What do these Jira board columns mean: To Do, In Progress, Done, Backlog?

- **To Do:** Tasks that are selected for the current sprint but have not been started yet.
- **In Progress:** Tasks that are currently being worked on by the team.
- **Done:** Tasks that have been completed, tested, and approved.
- **Backlog:** A list of all tasks, features, and requirements that need to be completed in the project.